

Hypothesis one sample t-tests



- 1 A researcher decides to test $H_0 : \mu = 23.5$ against $H_1 : \mu < 23.5$. The sample size of 45 is found to have a mean of 22.9 and a standard deviation of 1.2.

- a Find the value of the test statistic, t , correct to two decimal places.
- b Complete the statement:

$$p\text{-value} = P(T \leq \text{ }) , T \sim t_{n-1}.$$

- c Find the p -value for the sample, correct to four decimal places.
- d What does the p -value for the sample indicate about the null hypothesis. Is H_0 likely to be rejected or not rejected?

- 2 A researcher decides to test $H_0 : \mu = 55$ against $H_1 : \mu > 55$. The sample size of 60 is found to have a mean of 56.2 and a standard deviation of 4.2.

- a Find the value of the test statistic, t , correct to two decimal places.
- b Complete the statement:

$$p\text{-value} = P(T \geq \text{ }) , T \sim t_{n-1}.$$

- c Find the p -value for the sample, correct to four decimal places.
- d If the significance level is $\alpha = 5\%$, will H_0 be rejected or not rejected?

- 3 A researcher decides to test $H_0 : \mu = 16$ against $H_1 : \mu \neq 16$. The sample size of 30 is found to have a mean of 15.9 and a standard deviation of 1.36.

- a Find the value of the test statistic, t , correct to two decimal places.
- b Complete the statement:

$$p\text{-value} = P(T > \text{ }) \text{ or } p\text{-value} = P(T < \text{ }) , T \sim t_{n-1}.$$

- c Find the p -value for the sample, correct to four decimal places.
- d If the significance level is $\alpha = 10\%$, will H_0 be rejected or not rejected?



Applications

- 4 The company Speedy manufactures remote controlled cars, some of which come off the assembly line defective. The manufactures believe that the proportion of defective cars coming off their assembly line is about 6%, but a recent random sample of 100 cars contained 8 defective ones with a standard deviation of 2.37. They wonder if the proportion of defective cars is actually higher than the 6%.
- Define the set of hypotheses for this situation.
 - Use your calculator to find the value of the t -statistic, correct to three significant figures.
 - Calculate the p -value for the sample, correct to four decimal places.
 - If the significance level is $\alpha = 10\%$, does there appear to be evidence to justify Speedy's claim? Explain your answer.
- 5 The school bookshop tells prospective DP students that the average cost of its Mathematics textbooks is \$52. A group of smart statistics students think that the average cost is higher. In order to test the bookshop's claim, the students select a random sample of 100 books and find that the mean from their random sample is \$52.80 with a standard deviation of \$4.50.
- Define the hypothesis set for this problem.
 - Calculate the t -statistic for this one-tailed hypothesis problem, correct to two decimal places.
 - Calculate the p -value for the sample, correct to four decimal places.
 - If the significance level is $\alpha = 5\%$, will H_0 be rejected or not rejected?
- 6 A certain chemical pollutant in the Swan River has been constant for several years with mean $\mu = 34$ ppm (parts per million). A group of factory representatives whose companies discharge liquids into the river is now claiming that they have lowered the average with improved filtration devices. A group of environmentalists test to see if this is true. Their sample of size 50 gives a mean $\bar{x} = 32.5$ ppm and standard deviation $s = 8$ ppm.
- Define the hypothesis set for this problem.
 - Calculate the t -statistic for this one-tailed hypothesis problem, correct to two decimal places.
 - Calculate the p -value for the sample, correct to four decimal places.
 - If the significance level is $\alpha = 5\%$, does it appear that the factory representatives are correct in their claim? Explain your answer.

- 7 Based on population figures and other general information on the Australian population, on average, a family of four in Australia spends about \$1135 annually on dental expenditures. A regional dental association suspects that this amount is not correct and wants to test if this figure is accurate for their area of country. To test this, 20 families of 4 are randomly selected from the population in that area of the country and a log is kept of the family's dental expenditure for one year. The resulting data are given below. Assume that dental expenditure is normally distributed in the population.

1217, 1333, 1219, 1515, 799, 1121, 1187, 872, 932, 810,
689, 912, 933, 939, 964, 979, 1595, 989, 1002, 984

- Define the hypothesis set for this problem.
 - Find the t -statistic for this two-tailed hypothesis problem, correct to two decimal places.
 - Find the p -value for the sample, correct to four decimal places.
 - If the significance level is $\alpha = 10\%$, does it appear that the regional dental association is correct in its claim? Explain your answer.
- 8 The contents of 25 cans of soft drink labelled as 300 ml were measured and results were as follows:

301, 298, 289, 302, 299, 295, 300, 301, 297, 298, 300, 301,
302, 299, 295, 303, 297, 301, 300, 302, 302, 299, 298, 296, 302

Quality control wants to check that can contents are accurate within a 1% significance level. Assume that can contents data is normally distributed.

- Define the hypothesis set for this problem.
- Find the p -value for the sample, correct to two decimal places.
- State the sample mean, correct to two decimal places.
- Considering the significance level does it appear that there is a problem with the can contents for this brand? Explain your answer.